

Adaptive Memory Dynamics in Machine Learning Systems

Controlled forgetting, passive decay, and reinforcement-sensitive retention

Tier 1 - Capstone Research Concept, revised against implemented Stage One

Adaptive Memory Dynamics Research Program Stage One revision - September 2026

Evidence status: Stage One is a reproducible sklearn-digits-8x8 experiment across five independent seeds. Reinforced retention approximately matches the persistent control on ordinary accuracy and Task B adaptation; passive decay reduces accuracy; neither decay mechanism improves Task A retention in the current sequential setup.

Abstract

This study asks whether structured forgetting can be treated as an experimentally controllable learning mechanism rather than only as a failure mode. The implemented Stage One laboratory compares persistent retention, passive decay, and reinforcement-sensitive retention under identical model/training settings on sklearn digits. Five independent seeds provide a first frozen evidence bundle. The result is mixed: reinforced retention preserves ordinary test performance and Task B adaptation near the persistent control, passive decay lowers accuracy, and both decay mechanisms forget more of Task A in the current sequential experiment. The evidence therefore rejects any present claim that the implemented forgetting rules solve catastrophic forgetting while motivating controlled rate and reinforcement-strength ablations.

1. Research question

How do passive decay and reinforcement-sensitive retention alter generalization, post-idle stability, adaptation, and continual-learning retention relative to persistent parameter retention?

2. Mechanisms

Condition	Interpretation
Persistent baseline	No adaptive forgetting; control condition.
Passive decay	Underused representations weaken over time.
Reinforced retention	Frequently activated representations receive stronger retention weighting.

3. Frozen Stage One evidence

Mechanism	Final test	Post-idle	Task B	Task A forgetting
Persistent	0.9618 ± 0.0092	0.9618 ± 0.0092	0.9741 ± 0.0037	0.6947 ± 0.1014
Passive	0.9480 ± 0.0088	0.9458 ± 0.0051	0.9661 ± 0.0051	0.9858 ± 0.0058
Reinforced	0.9622 ± 0.0047	0.9613 ± 0.0058	0.9759 ± 0.0024	0.8619 ± 0.0558

Figure 1. Measured mean accuracy $\pm$ sample SD across five independent Stage One seeds.

Figure 2. Measured Task A forgetting $\pm$ sample SD. Lower is better; the persistent baseline retained Task A best in this configuration.

4. Revised hypothesis discipline

- H1: Memory mechanism changes at least one retention/adaptation outcome under controlled training.
- Reinforced retention may preserve ordinary accuracy while altering sequential retention; this is a testable tradeoff, not an assumed benefit.
- An adaptive forgetting mechanism is considered beneficial only when gains on a pre-specified adaptation/generalization outcome are not purchased through unacceptable retention loss.
- Null and negative outcomes are valid and retained.

5. Claim boundary

Stage One validates the experiment/export pipeline on sklearn-digits-8x8. It does not establish performance on MNIST, Fashion-MNIST, CIFAR-10, broad continual-learning benchmarks, or biological fidelity. It also does not show that controlled forgetting improves catastrophic forgetting.

Figures

